

John Kelliher

Senior Software Engineer

jackwkelliher@gmail.com | GitHub: jack-work | john.kelliher.info

PROFESSIONAL SUMMARY

Senior Software Engineer at Microsoft building distributed infrastructure for agentic development platforms and real-time collaborative state systems at global scale. End-to-end engineer spanning backend architecture, cloud deployment, observability, container orchestration, and Linux systems development — equally at home designing persistent data structures, wiring deployments, or fighting livesite fires with panache. Compiler design background. Deployed across 30+ regions including sovereign and air-gapped clouds, serving products within the Microsoft 365 Copilot ecosystem (100M+ MAU) and Power Platform (56M+ MAU).

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Microsoft — Power Platform & M365 Copilot

July 2019 – Present

Local-First Replicated Coding Agent Infrastructure (Fall 2025 – Present)

- Architected the backend of an event-sourced, local-first replicated state service on the virtual actor model (Orleans), serving as the state kernel for Power Apps and M365 Copilot products — enabling real-time collaborative editing for both human developers and autonomous agents
- Designed a persistent immutable trie with structural sharing for versioned state, synchronized via server-authoritative optimistic concurrency over WebSocket and HTTP; client replicas apply and merge structural and text-based patches with server-arbitrated conflict resolution
- Engineered a write-ahead log with segmented compaction and snapshot-based watermarking for recovery, backed by hot-tier Redis and tiered object storage; designed log compaction with consistency reconciliation through scalable job systems
- Built FUSE virtual filesystem with write-through local caching and debounced network propagation, providing agents a native filesystem interface to replicated state; materializes Git-compatible linear change history via virtual .git for a familiar developer workflow
- Delivered first-party client libraries (TypeScript, C#) and React shell implementing local-first replication with optimistic writes and rollback; code-generated schema bindings for both filesystem-native (agentic) and Power Apps state models
- Engineered container infrastructure with Linux namespace isolation, VNet egress control, and systemd-managed agent processes; supports multiple harness protocols including MCP, with on-container and off-container agent execution via fork/exec and network proxy
- Deployed globally across 30+ regions and sovereign clouds using Bicep, ARM, and Ev2 rollouts orchestrated via Azure DevOps; built observability stack on Geneva and Grafana with time-series telemetry, distributed tracing, and alerting conforming to Microsoft safe deployment practices

Compiler & Platform Engineering (2019 – 2025)

- Co-led three-person senior team to rewrite the Power Apps formula compiler, achieving 500% throughput improvement and orders-of-magnitude gains on pathological inputs
- Secured enterprise funding for an experimental AI-powered development platform with sandboxed execution; built novel LLM streaming framework that evolved into current agentic infrastructure
- Architected telemetry platform on OpenTelemetry and built distributed conflict-free data structures for real-time collaboration
- Designed authentication middleware enabling entry into compliance-heavy enterprise markets while reducing operational costs

Software Engineering Intern | Microsoft

Summer 2018

- Built interactive debugging system for Power Apps that became the production-standard development tool

Software Intern | Hankook Tire

2016 – 2017

- Developed internal systems, employee management platforms, and public-facing website

EDUCATION

Master of Science in Computer Science, Specialization in AI Systems

Fall 2025 – Present

Georgia Institute of Technology

Bachelor of Science in Computer Science and Engineering

2015 – 2019

The Ohio State University

TECHNICAL PROFICIENCIES

Programming	Infrastructure	Architecture	Specialized
C# and .NET, Go, TypeScript, Python, React, Linux/Unix	Azure, Bicep, ARM, Docker, Redis, Terraform, Service Fabric, KQL	Event sourcing, actor model, local-first replication, microservices, WebSocket/HTTP	Agentic systems (MCP), FUSE, systemd/dbus, compiler design, OpenTelemetry, Geneva/Grafana

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